Autonomous Vehicle Competition

ROBO 5302 / CSCI 4302/5302: Advanced Robotics

For your final project, you will implement an autonomous vehicle controller that can navigate the DeepRacer through a closed indoor course. The Autonomous Vehicle Competition will be divided into two parts: a **time trial race** around the course and the implementation of **challenge features**. The grading for the project is split as follows:

- **Hardware Demonstrations** throughout the semester (4 x 3.75% each) – 15%
- **Project Proposal** (due Fri 4/3) – 5%
- **Average lap time** around the course – 25%
- Completion of **challenge features** – 45%
- Project Report – 10%

The final competition will take place during the final exam time, currently scheduled for Fri 1 May at 10:30am - 1pm. All project deliverables (e.g. the report, code, and feature demonstration videos) are due prior to the start of the competition and **all students are required to be present with no exceptions**. **Individuals not present for the final demonstration may be deducted a significant portion of the group's project grade.** Code should be submitted as a link to a GitHub Repo with your ROS packages.

Minimum Challenge Feature Requirements Per Team

Graduate Teams (Clanker Collective, The Avengers, The Copperbottoms, The Corner)

- Total Challenge **Points**: 8
- Number of 8 point **Challenges**: 1

Undergraduate Teams (Underqualified)

- Total Challenge **Points**: 5

You may complete as many challenges as you like, with extra credit being awarded for additional points exceeding the required minimum. The amount of extra credit will be awarded at the discretion of the professor, with a maximum allowable Final Project grade of 110%. Extra credit will only be awarded if a minimum grade of 90% is achieved for all problems (base features AND extra credit features), ie. you cannot attempt many problems and receive a low grade for all of them to ultimately generate a high cumulative final grade.

Project Proposal

Each group is required to write a 1-2 page project proposal due on Fri 4/3. In this proposal you need to clearly indicate which challenge problems you plan to attempt and the total number of points you plan to score. You are required to explain briefly how you will approach the challenge features including addressing compute, sensor integration and algorithm selection (please include citations/links to online sources/packages). Also include your proposed approach to autonomously complete the course. **It is critical that you also discuss what concerns you have regarding your project and how you plan to**

mitigate those risks. If your group chooses to divvy up components of your project, please include which group members will be responsible for different tasks.

Technical Writeup

All teams are required to turn in a report between 6 and 8 pages in ICRA format that includes a description of their system, implementation details, and performance analysis for time trial, and all completed challenges. The report should be written for a technical audience who is not familiar with the challenge, such as a prospective future employer. The page limit does not include appendices (do not include code in your appendix but rather include a link to your repo); however, the reader should not need to reference the appendices to understand what you have created. The appendix should include which challenge features you attempted for the final project submission. This means important figures should be included in the main report, with supporting figures and documentation available in the appendices for interested parties. Reports are to be completed using Latex. A template for ICRA format can be found at the following link: **http://ras.papercept.net/conferences/support/tex.php.**

Scoring

For the time trial portion of the competition, teams are to run three timed laps during the final competition day. These laps are completely separate from the challenge features below. The race course will be around a loop of the basement of the engineering center in a clockwise loop (right turns only). The team with the overall fastest average lap will be crowned champions of the event and will gain a **10% bonus** for the time trial portion of your grade (and, more importantly, bragging rights among your peers). The team with the lowest variance in lap time (most consistent) will receive a **10% bonus** for the time trial portion of your grade. All teams that can achieve average lap times below a threshold of 2 minutes and 30 seconds (3 minutes for undergraduate groups) will receive 100% for the time trial portion of the challenge. Times above the threshold will receive a grade on a linear sliding scale. Collisions with any obstacle will result in a 5% point reduction and a required human intervention will result in a 10% point reduction. Ultimately, your vehicle should be able to **reliably** complete the course.

Challenge feature implementations are graded based on capability and reliability – your implementations are expected to work multiple attempts in a row and not just every now and then. Partial implementations of challenges will be graded according to how far along the implementation is (as opposed to "all or nothing" grading) – in these cases, please elaborate in your report about what isn't working.

Participation

Your report appendix must include a detailed description of each team member's contribution. **Separately, there will be an anonymous feedback form that must be completed by everyone indicating relative contributions of each team member. This will be taken into consideration for individual grading.**

Use of Existing Libraries

We have explicitly stated in the challenges below when it is permissible to use existing libraries. Assume all modules must be written from scratch by your group unless explicitly stated otherwise. Common packages like OpenCV or NumPy are acceptable. When in doubt please post a question on Piazza.

Hardware Return

Hardware will be returned immediately following the competition. Be sure to bring all checked out equipment. **Final grades will not be assigned to groups if ALL hardware is not returned neatly organized in the boxes provided. Additionally, groups will be required to show that all components are fully functioning upon return.**

8-Point Challenges:

- **LIDAR-based EKF for Localization**:
    - Implement an EKF on your platform for localization. The EKF should predict the vehicle's pose using a vehicle kinematics model and a vehicle sensor model.
    - You will need to collect data ahead of time to represent the world. You then need to show how your algorithm performs at localizing within this existing map you created.
    - If you choose to use any scan matching algorithms (such as ICP), they may be drawn from online sources. Your overall localization needs to be implemented from scratch though.
- **Landmark based SLAM**
    - Landmark based SLAM for localization mapping. Teams can make their own landmarks to place around the course and use the landmarks to localize. Reminder that the demos are taking place in a public space, so please do not create landmarks that inhibit the use of this space.
    - Teams are required to be able to produce a map within which their robot can localize. Live visualization of this is required in a package such as Rviz.
    - Online repos are allowed for object recognition.
    - No online repos are allowed for the SLAM implementation.
- **SLAM package implementation**
    - Implement a two-mode system. In "Mapping Mode" the system runs a SLAM algorithm to localize and build a map. In "Planning Mode" the system will freeze the map, generate a path over your map and use your SLAM solution only to localize (because the map has already been constructed in the mapping phase). Planning mode can be used in conjunction with a planning algorithm and low-level controller of your choice to navigate the course for your timed laps. You may use this implementation for your time trial.
    - The SLAM package can be from an online GitHub repository. Planning and low-level control needs to be completed from scratch.

3-Point Challenges

- **Parallel Parking:**
    - Create a controller that drives your vehicle forward until it detects a parking spot on the right side. Your controller should identify the opening and successfully (collision free) park your car between in a rectangular "parking spot". The setup will be provided to groups ahead of time to help with algorithm tuning.
- **Camera Navigation**
    - Using only the onboard RGB camera(s), leverage cues in the environment to indicate when your vehicle should execute turns to quickly navigate around the course. Dark-colored tape will be laid on the floor down the center of the course.

    - No Online Repos can be used for this. OpenCV may not be used for this problem.
- **Reverse Driving:**
    - Using any of the available sensors on the vehicle, complete the course going backwards.

2-Point Challenges:

- **Avoid an obstacle in the track (during a lap of the course):**
    - Obstacle(s) will be placed in the course during your lap. Your car should avoid hitting the obstacle while continuing to advance through the course. Your write-up should include figures demonstrating how the obstacle is identified and how the planning avoids it.
    - This is an additional lap of the course; it is not counted toward your time trials.

1-Point Challenges:

- Add **stop signs** to the environment and obey them with your controller: bringing the vehicle to a complete stop, pausing for a moment, then continuing. (make sure this can be turned off with a Boolean flag in your controller for measuring lap times)
- **Implement two different control approaches** and compare them in a dedicated section of your project writeup. Discuss why you chose one over the other in your final controller.
- Build and display an **interface to visualize vehicle telemetry** (speed, steering angle, other relevant sensor information) as your controller runs. Rviz (or any other robot visualization package) may not be used for this feature. Matplotlib is acceptable.